

Kaustubh Rathi

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SUMMARY

Applied Generative AI engineer building production LLM-agent systems at Amazon — Amazon Bedrock, multi-step agentic orchestration, tool use, and RAG — with strong Python/Java/C++ and algorithms fundamentals. Ships agents that run reliably in production, not just prototypes.

EDUCATION

B.Tech in Information Technology (Minor in Economics) 2022 – 2026
Indian Institute of Information Technology (IIIT), Allahabad — CGPA: 8.52/10 Prayagraj, India

TECHNICAL SKILLS

GenAI & LLM: LLM agents, multi-agent orchestration, tool use, Amazon Bedrock, RAG, embeddings, vector search (FAISS), knowledge graphs, prompt engineering, LangChain, LangGraph, NER
ML: PyTorch, TensorFlow, Transformers, Hugging Face, scikit-learn
Cloud & Serving: AWS (Bedrock, Lambda, DynamoDB, CloudWatch), Docker
Languages: Python, Java, C++, SQL

EXPERIENCE

Software Development Engineer Intern Jan 2026 – Present
Amazon — Grocery Ordering, Stores Bengaluru, India

- Architected an autonomous order-failure remediation agent on Amazon Bedrock: a multi-call LLM pipeline that investigates ~150 failed customer orders/day (verify execution → search tickets/code → synthesize root cause), semantically clusters orders sharing a cause, and auto-generates code-fix CRs to prevent recurrence.
- Engineered the agent's reliability layer: a crash-resilient state machine with per-call output validation, automatic retries, 1–5 confidence scoring, and human-in-the-loop approval gates before any code or data change.
- Shipped a 24/7 incident-investigation agent (Bedrock + tool use) end to end that scans tickets every 5 minutes and correlates CloudWatch/Lambda/Step Functions evidence with deployments to auto-generate 5-level root-cause analyses, cutting investigation time from ~2 hours to ~15 minutes; including the supporting Java backend and knowledge-base data sources.

Artificial Intelligence Intern Dec 2024 – Feb 2025
ASPR Agrovalue (I) Limited India

- Automated an invoice-processing pipeline using NER to extract structured fields from 10,000+ supplier invoices/month into a normalized schema, replacing manual data entry.
- Developed a Python web-scraping + ETL pipeline aggregating 1,000+ price/market records/day from fragmented sources, and a Streamlit dashboard the procurement team adopted for price-trend visibility.

PROJECTS

Deep Researcher | *Python, LangGraph, FastAPI, Flask, WebSocket, FAISS* 🐙 GitHub

- Built a multi-agent research-automation web app orchestrating 7 specialized LLM agents (research, edit, write, review, revise, publish) via LangGraph with a reviewer-reviser loop, producing 10–15 page cited reports in ~5 minutes; FastAPI backend with WebSocket log streaming and PDF/DOCX/Markdown export.

Medical Chatbot | *LangChain, FAISS, OpenAI, Gradio, Docker* 🐙 GitHub

- Created a Dockerized RAG clinical-interview chatbot supporting 6 languages (including Hindi) that conducts structured 10–25 question interviews and generates assessment reports.
- Grounded responses in a FAISS vector store of ~5,000 real doctor-patient dialogue chunks, with persona-driven prompting that improved clinical relevance over an ungrounded baseline in manual evaluation.

Knowledge-Graph RAG | *Neo4j, Cypher, embeddings, LLMs* 🐙 GitHub

- Designed a graph-grounded Q&A chatbot over 100+ SEC 10-K filings using a Neo4j knowledge graph (Cypher), embeddings, and entity/relation extraction for retrieval-augmented answers.
- Benchmarked graph-based retrieval against plain vector RAG, showing relationship-aware traversal raised answer accuracy ~20% on multi-hop financial questions.

ACHIEVEMENTS & LEADERSHIP

- Codeforces Expert** (max rating 1751) | **LeetCode Knight** (rating 2000+), 800+ problems solved.
- Secured **99.45 percentile** in JEE Mains 2022 (top 0.55% among 1M+ candidates).
- Overall Coordinator, Robita Club (Generative AI), IIIT Allahabad** (2023–25): led 10+ students, mentored 100+ juniors via AI hackathons and PyTorch workshops.